

Unit 9. Positive (Semi)definite Matrices

1 Quadratic forms and factorisations

1.1 Definiteness and congruence

Definition 9.1 (Quadratic form and congruence). For $\mathbf{A} \in \mathbb{R}^{n \times n}$, the function

$$q_{\mathbf{A}}(\mathbf{x}) = \mathbf{x}'\mathbf{A}\mathbf{x}, \quad \mathbf{x} \in \mathbb{R}^n,$$

is the **quadratic form represented by $\mathbf{A}$** . Symmetric matrices $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$ are **congruent** if

$$\mathbf{B} = \mathbf{P}'\mathbf{A}\mathbf{P}$$

for some nonsingular $\mathbf{P} \in \mathbb{R}^{n \times n}$. Under the coordinate change $\mathbf{x} = \mathbf{P}\mathbf{y}$,

$$q_{\mathbf{A}}(\mathbf{x}) = q_{\mathbf{P}'\mathbf{A}\mathbf{P}}(\mathbf{y}).$$

Origin: H §14.1; S §1.3.4

Lemma 9.2 (The symmetric representative). For every $\mathbf{A} \in \mathbb{R}^{n \times n}$,

$$q_{\mathbf{A}} = q_{(\mathbf{A} + \mathbf{A}')/2}.$$

If $\mathbf{S}$ and $\mathbf{T}$ are symmetric, then

$$q_{\mathbf{S}}(\mathbf{x}) = q_{\mathbf{T}}(\mathbf{x}) \quad (\mathbf{x} \in \mathbb{R}^n) \iff \mathbf{S} = \mathbf{T}.$$

Origin: A&M Exs. 8.1 and 8.3; H Lemma 14.1.1 and Cor. 14.1.3

Proof. The scalar $\mathbf{x}'\mathbf{A}\mathbf{x}$ equals its transpose, so

$$\mathbf{x}'\mathbf{A}\mathbf{x} = \frac{1}{2}\{\mathbf{x}'\mathbf{A}\mathbf{x} + \mathbf{x}'\mathbf{A}'\mathbf{x}\} = \mathbf{x}'\frac{\mathbf{A} + \mathbf{A}'}{2}\mathbf{x}.$$

Let $\mathbf{D} = \mathbf{S} - \mathbf{T} = (d_{ij})$. If $\mathbf{x}'\mathbf{D}\mathbf{x} = 0$ for every $\mathbf{x}$, then

$$d_{ii} = \mathbf{e}_i'\mathbf{D}\mathbf{e}_i = 0,$$

and, for $i \neq j$,

$$0 = (\mathbf{e}_i + \mathbf{e}_j)'\mathbf{D}(\mathbf{e}_i + \mathbf{e}_j) = d_{ii} + 2d_{ij} + d_{jj} = 2d_{ij}.$$

Thus $\mathbf{D} = \mathbf{O}$. If $\mathbf{S} = \mathbf{T}$, their quadratic forms are equal. □

Definition 9.3 (Definiteness). Let $\mathbf{A} = \mathbf{A}' \in \mathbb{R}^{n \times n}$.

$$\mathbf{A} \succeq \mathbf{O} \iff \mathbf{x}'\mathbf{A}\mathbf{x} \geq 0 \quad (\mathbf{x} \in \mathbb{R}^n),$$

$$\mathbf{A} \succ \mathbf{O} \iff \mathbf{x}'\mathbf{A}\mathbf{x} > 0 \quad (\mathbf{0} \neq \mathbf{x} \in \mathbb{R}^n).$$

The first condition is **positive semidefiniteness**; the second is **positive definiteness**. The matrix is **indefinite** if its quadratic form takes both positive and negative values.

Origin: A&M §8.1 opening; H §14.2a; S §1.3.5

Theorem 9.4 (Spectral and Gram characterisations). Let $\mathbf{A} = \mathbf{A}' \in \mathbb{R}^{n \times n}$, with eigenvalues $\lambda_1, \dots, \lambda_n$. The following are equivalent:

$$\mathbf{A} \succeq \mathbf{O}; \quad \lambda_i \geq 0 \ (i = 1, \dots, n); \quad \mathbf{A} = \mathbf{G}'\mathbf{G} \text{ for some } \mathbf{G} \in \mathbb{R}^{n \times n}.$$

Under these conditions,

$$\mathbf{x}'\mathbf{A}\mathbf{x} = 0 \iff \mathbf{A}\mathbf{x} = \mathbf{0}, \quad \text{rank}(\mathbf{A}) = \text{rank}(\mathbf{G})$$

for the spectral Gram factor constructed in the proof. Moreover,

$$\mathbf{A} \succ \mathbf{0} \iff \lambda_i > 0 \ (i = 1, \dots, n) \iff \mathbf{A} \succeq \mathbf{0} \text{ and } \mathbf{A} \text{ is nonsingular.}$$

Origin: A&M Exs. 8.10–8.14 and 8.21; H Thm. 14.3.7 and Thm. 21.8.5

Proof. By the real spectral theorem,

$$\mathbf{A} = \mathbf{Q} \operatorname{diag}(\lambda_1, \dots, \lambda_n) \mathbf{Q}', \quad \mathbf{Q}' \mathbf{Q} = \mathbf{I}_n.$$

For $\mathbf{y} = \mathbf{Q}' \mathbf{x}$,

$$\mathbf{x}' \mathbf{A} \mathbf{x} = \sum_{i=1}^n \lambda_i y_i^2.$$

This is nonnegative for every $\mathbf{y}$ exactly when every $\lambda_i \geq 0$, and positive for every $\mathbf{y} \neq \mathbf{0}$ exactly when every $\lambda_i > 0$. When $\lambda_i \geq 0$, set

$$\mathbf{G} = \operatorname{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n}) \mathbf{Q}'.$$

Then $\mathbf{G}'\mathbf{G} = \mathbf{A}$. Conversely, $\mathbf{A} = \mathbf{G}'\mathbf{G}$ gives $\mathbf{x}'\mathbf{A}\mathbf{x} = \|\mathbf{G}\mathbf{x}\|^2 \geq 0$. For the displayed $\mathbf{G}$,

$$\mathbf{x}'\mathbf{A}\mathbf{x} = 0 \iff \sqrt{\lambda_i}y_i = 0 \ (i = 1, \dots, n) \iff \lambda_i y_i = 0 \ (i = 1, \dots, n) \iff \mathbf{A}\mathbf{x} = \mathbf{0}.$$

Both $\text{rank}(\mathbf{A})$ and $\text{rank}(\mathbf{G})$ equal the number of nonzero λ_i . Finally, all $\lambda_i > 0$ exactly when they are all nonnegative and none is zero, which is equivalent to nonsingularity. $\square$

Theorem 9.5 (Rectangular congruence transforms). Let $\mathbf{A} = \mathbf{A}' \in \mathbb{R}^{n \times n}$ and $\mathbf{P} \in \mathbb{R}^{n \times m}$.

$$\mathbf{A} \succeq \mathbf{0} \implies \mathbf{P}'\mathbf{A}\mathbf{P} \succeq \mathbf{0}.$$

If $\mathbf{A} \succ \mathbf{0}$, then

$$\mathbf{P}'\mathbf{A}\mathbf{P} \succ \mathbf{0} \iff \text{rank}(\mathbf{P}) = m.$$

Consequently, for square nonsingular $\mathbf{P}$,

$$\mathbf{P}'\mathbf{A}\mathbf{P} \succeq \mathbf{0} \iff \mathbf{A} \succeq \mathbf{0}, \quad \mathbf{P}'\mathbf{A}\mathbf{P} \succ \mathbf{0} \iff \mathbf{A} \succ \mathbf{0}.$$

Origin: A&M Exs. 8.26 and 8.28; H Thm. 14.2.9 and Cor. 14.2.10

Proof. For $\mathbf{x} \in \mathbb{R}^m$,

$$\mathbf{x}'\mathbf{P}'\mathbf{A}\mathbf{P}\mathbf{x} = (\mathbf{P}\mathbf{x})'\mathbf{A}(\mathbf{P}\mathbf{x}).$$

This proves the semidefinite implication. If $\mathbf{A} \succ \mathbf{O}$, the last quantity is positive for every $\mathbf{x} \neq \mathbf{0}$ exactly when $\ker(\mathbf{P}) = \{\mathbf{0}\}$, equivalently $\text{rank}(\mathbf{P}) = m$. For square nonsingular $\mathbf{P}$, apply the first implications to $\mathbf{P}$ and to $\mathbf{P}^{-1}$, using

$$\mathbf{A} = (\mathbf{P}^{-1})'(\mathbf{P}'\mathbf{A}\mathbf{P})\mathbf{P}^{-1}.$$

□

Lemma 9.6 (Entry restrictions). If $\mathbf{A} = (a_{ij}) = \mathbf{A}' \succeq \mathbf{O}$, then

$$a_{ii} \geq 0, \quad a_{ij}^2 \leq a_{ii}a_{jj}.$$

If $a_{ii} = 0$, then row i and column i are zero.

Origin: A&M Ex. 8.7; H Cor. 14.5.14

Proof. First, $a_{ii} = \mathbf{e}_i' \mathbf{A} \mathbf{e}_i \geq 0$. Fix $i \neq j$. If $a_{ii} > 0$, then

$$0 \leq \left(-\frac{a_{ij}}{a_{ii}} \mathbf{e}_i + \mathbf{e}_j \right)' \mathbf{A} \left(-\frac{a_{ij}}{a_{ii}} \mathbf{e}_i + \mathbf{e}_j \right) = a_{jj} - \frac{a_{ij}^2}{a_{ii}}.$$

If $a_{ii} = 0$ and $a_{ij} \neq 0$, take $t = -(a_{jj} + 1)/(2a_{ij})$. Then

$$(t\mathbf{e}_i + \mathbf{e}_j)' \mathbf{A} (t\mathbf{e}_i + \mathbf{e}_j) = 2a_{ij}t + a_{jj} = -1,$$

a contradiction. Thus $a_{ij} = 0$ for every j , and symmetry also gives the zero column. The stated inequality then holds in the remaining case. $\square$

Example 9.1 (Three signs). For $\mathbf{x} = (x, y)'$,

$\mathbf{A}$	$\mathbf{x}'\mathbf{A}\mathbf{x}$	classification
$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$	$(x + y)^2 + x^2 + y^2$	$\mathbf{A} \succ \mathbf{O}$
$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$	$(x + y)^2$	$\mathbf{A} \succeq \mathbf{O}$ and singular
$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	$x^2 - y^2$	indefinite.

Origin: Adapted from A&M Ex. 8.2 and H §14.2a

1.2 Leading principal minors

Definition 9.7 (Principal and leading principal minors). For $I \subseteq \{1, \dots, n\}$, the matrix $\mathbf{A}[I, I]$ is the **principal submatrix indexed by I** , and $\det \mathbf{A}[I, I]$ is its **principal minor**. For

$I = \{1, \dots, k\}$, write

$$\mathbf{A}_k = \mathbf{A}[1:k, 1:k], \quad \Delta_k = \det(\mathbf{A}_k), \quad \Delta_0 = 1;$$

these are the **leading** principal submatrices and minors.

Origin: A&M Exs. 8.30–8.32; H §14.9c

Theorem 9.8 (Sylvester's criterion). For $\mathbf{A} = \mathbf{A}' \in \mathbb{R}^{n \times n}$, $n \geq 1$,

$$\mathbf{A} \succ \mathbf{O} \iff \Delta_k > 0 \quad (k = 1, \dots, n).$$

Nonnegativity of the leading principal minors alone does not characterise positive semidefiniteness.

Origin: A&M Exs. 8.31–8.33; H Thm. 14.9.5

Proof. If $\mathbf{A} \succ \mathbf{O}$, set $\mathbf{P}_k = (\mathbf{e}_1, \dots, \mathbf{e}_k)$. Then

$$\mathbf{A}_k = \mathbf{P}'_k \mathbf{A} \mathbf{P}_k, \quad \text{rank}(\mathbf{P}_k) = k.$$

Theorem 9.5 gives $\mathbf{A}_k \succ \mathbf{O}$. Its eigenvalues are positive, so $\Delta_k > 0$.

Conversely, suppose every $\Delta_k > 0$. The leading-principal-minor LDU result gives a unique factorisation

$$\mathbf{A} = \mathbf{L}\mathbf{D}\mathbf{U}, \quad \mathbf{D} = \text{diag}(d_1, \dots, d_n), \quad d_k = \frac{\Delta_k}{\Delta_{k-1}} > 0.$$

Since $\mathbf{A} = \mathbf{A}' = \mathbf{U}'\mathbf{D}\mathbf{L}'$, uniqueness gives $\mathbf{U} = \mathbf{L}'$. For $\mathbf{x} \neq \mathbf{0}$, $\mathbf{L}'\mathbf{x} \neq \mathbf{0}$, and

$$\mathbf{x}'\mathbf{A}\mathbf{x} = (\mathbf{L}'\mathbf{x})'\mathbf{D}(\mathbf{L}'\mathbf{x}) > 0.$$

The matrix

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

has leading principal minors 0, 0, 1, but its quadratic form takes both signs. □

1.3 Cholesky factorisations

Theorem 9.9 (Positive definite Cholesky factorisation). If $\mathbf{A} = \mathbf{A}' \succ \mathbf{O}$, there is a unique lower triangular matrix $\mathbf{L}$ with positive diagonal such that

$$\mathbf{A} = \mathbf{L}\mathbf{L}'.$$

Moreover,

$$\det(\mathbf{A}) = \left(\prod_{i=1}^n \ell_{ii} \right)^2.$$

Origin: A&M Ex. 8.23; H Thm. 14.5.11; S Thm. 11.2

Proof. The proof of Sylvester's criterion gives

$$\mathbf{A} = \mathbf{C}\mathbf{D}\mathbf{C}', \quad \mathbf{C} \text{ unit lower triangular}, \quad \mathbf{D} = \text{diag}(d_1, \dots, d_n), \quad d_i > 0.$$

Thus $\mathbf{L} = \mathbf{C} \text{diag}(\sqrt{d_1}, \dots, \sqrt{d_n})$ has the required properties. If also $\mathbf{A} = \tilde{\mathbf{L}}\tilde{\mathbf{L}}'$, set $\mathbf{T} = \tilde{\mathbf{L}}^{-1}\mathbf{L}$.

Then $\mathbf{T}$ is lower triangular with positive diagonal and

$$\mathbf{T}\mathbf{T}' = \mathbf{I}_n, \quad \mathbf{T}' = \mathbf{T}^{-1}.$$

The left side of the second equality is upper triangular and the right side is lower triangular; hence $\mathbf{T}$ is diagonal. Its positive diagonal entries satisfy $t_{ii}^2 = 1$, so $\mathbf{T} = \mathbf{I}_n$ and $\tilde{\mathbf{L}} = \mathbf{L}$. Taking determinants gives the last identity. $\square$

Theorem 9.10 (Semidefinite Cholesky factorisation). If $\mathbf{A} = \mathbf{A}' \succeq \mathbf{O}$ has rank r , there is a unique lower triangular $\mathbf{L} = (\ell_{ij})$ such that

$$\mathbf{A} = \mathbf{L}\mathbf{L}', \quad \ell_{ii} \geq 0, \quad \ell_{ii} = 0 \implies \mathbf{L}\mathbf{e}_i = \mathbf{0}.$$

Exactly r diagonal entries are positive. For $\mathbf{A} = \mathbf{O}$, the factor is $\mathbf{L} = \mathbf{O}$.

Origin: A&M Ex. 8.23; H Thm. 14.5.16

Proof. Induct on n . For $n = 1$,

$$\mathbf{A} = (a_{11}), \quad \mathbf{L} = (\sqrt{a_{11}}).$$

For $n > 1$, write

$$\mathbf{A} = \begin{pmatrix} a & \mathbf{r}' \\ \mathbf{r} & \mathbf{B} \end{pmatrix}.$$

If $a = 0$, Lemma 9.6 gives $\mathbf{r} = \mathbf{0}$. Apply induction to $\mathbf{B}$ and place a zero first row and column in its factor; $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{B})$, so the diagonal count is preserved.

If $a > 0$, set $c = \sqrt{a}$, $\mathbf{v} = \mathbf{r}/c$, and $\mathbf{S} = \mathbf{B} - \mathbf{v}\mathbf{v}'$. For every $\mathbf{y}$, taking $t = -\mathbf{r}'\mathbf{y}/a$ gives

$$\mathbf{y}'\mathbf{S}\mathbf{y} = \begin{pmatrix} t \\ \mathbf{y} \end{pmatrix}' \mathbf{A} \begin{pmatrix} t \\ \mathbf{y} \end{pmatrix} \geq 0.$$

By induction, $\mathbf{S} = \mathbf{L}_*\mathbf{L}_*'$ with the stated convention. Then

$$\mathbf{L} = \begin{pmatrix} c & \mathbf{0}' \\ \mathbf{v} & \mathbf{L}_* \end{pmatrix}, \quad \mathbf{L}\mathbf{L}' = \mathbf{A}.$$

Also

$$\mathbf{A} = \begin{pmatrix} c & \mathbf{0}' \\ \mathbf{v} & \mathbf{I} \end{pmatrix} \begin{pmatrix} 1 & \mathbf{0}' \\ \mathbf{0} & \mathbf{S} \end{pmatrix} \begin{pmatrix} c & \mathbf{v}' \\ \mathbf{0} & \mathbf{I} \end{pmatrix},$$

so $\text{rank}(\mathbf{A}) = 1 + \text{rank}(\mathbf{S})$. Thus the number of positive diagonal entries equals r . At each induction step, a determines c ; if $c > 0$, $\mathbf{r}$ determines $\mathbf{v}$, and if $c = 0$, the null-column condition determines the first column. Induction gives uniqueness. $\square$

2 Square roots and matrix order

2.1 The symmetric square root

Theorem 9.11 (Positive semidefinite square root). For every $\mathbf{A} = \mathbf{A}' \succeq \mathbf{O}$, there is a unique symmetric positive semidefinite matrix $\mathbf{R}$ satisfying $\mathbf{R}^2 = \mathbf{A}$. If

$$\mathbf{A} = \mathbf{Q} \text{diag}(\lambda_1, \dots, \lambda_n) \mathbf{Q}', \quad \lambda_i \geq 0,$$

then

$$\mathbf{A}^{1/2} := \mathbf{R} = \mathbf{Q} \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n}) \mathbf{Q}'.$$

It is positive definite exactly when $\mathbf{A}$ is positive definite.

Origin: A&M Ex. 8.24; H Thm. 21.9.1; S Thm. 6.1

Proof. The displayed matrix is symmetric positive semidefinite and its square is $\mathbf{A}$. Suppose $\mathbf{S} = \mathbf{S}' \succeq \mathbf{O}$ and $\mathbf{S}^2 = \mathbf{A}$. Then

$$\mathbf{SA} = \mathbf{S}^3 = \mathbf{AS}.$$

Commuting symmetric matrices are simultaneously orthogonally diagonalizable. In a common eigenbasis, $\mathbf{S}^2 = \mathbf{A}$ becomes

$$s_i^2 = \lambda_i, \quad s_i \geq 0,$$

so $s_i = \sqrt{\lambda_i}$ for every i . Hence $\mathbf{S} = \mathbf{R}$. The square root has positive eigenvalues exactly when every $\sqrt{\lambda_i} > 0$, equivalently every $\lambda_i > 0$. $\square$

Corollary 9.12 (Inverse square root). If $\mathbf{A} \succ \mathbf{O}$, define

$$\mathbf{A}^{-1/2} = \mathbf{Q} \operatorname{diag}(\lambda_1^{-1/2}, \dots, \lambda_n^{-1/2}) \mathbf{Q}'.$$

Then

$$\mathbf{A}^{-1/2} = (\mathbf{A}^{1/2})^{-1} = (\mathbf{A}^{-1})^{1/2}, \quad \mathbf{A}^{-1/2} \mathbf{A} \mathbf{A}^{-1/2} = \mathbf{I}_n.$$

Origin: A&M Ex. 8.25; H Thm. 21.9.1

Proof. All four matrices are diagonal in the same orthonormal eigenbasis, and their diagonal entries satisfy

$$(\sqrt{\lambda_i})^{-1} = \lambda_i^{-1/2} = \sqrt{\lambda_i^{-1}}, \quad \lambda_i^{-1/2} \lambda_i \lambda_i^{-1/2} = 1.$$

□

Example 9.2 (Cholesky factor versus symmetric square root). For

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix},$$

the lower Cholesky factor and the symmetric square root are

$$\mathbf{L} = \begin{pmatrix} \sqrt{2} & 0 \\ 1/\sqrt{2} & \sqrt{3/2} \end{pmatrix}, \quad \mathbf{A}^{1/2} = \frac{1}{2} \begin{pmatrix} \sqrt{3} + 1 & \sqrt{3} - 1 \\ \sqrt{3} - 1 & \sqrt{3} + 1 \end{pmatrix}.$$

Thus $\mathbf{L}\mathbf{L}' = \mathbf{A} = (\mathbf{A}^{1/2})^2$, but $\mathbf{L} \neq \mathbf{A}^{1/2}$.

Origin: Course formulation from H Thm. 14.5.11 and Thm. 21.9.1

2.2 The Löwner order

Definition 9.13 (Löwner order). For symmetric $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$,

$$\mathbf{A} \succeq \mathbf{B} \iff \mathbf{A} - \mathbf{B} \succeq \mathbf{O}, \quad \mathbf{A} \succ \mathbf{B} \iff \mathbf{A} - \mathbf{B} \succ \mathbf{O}.$$

Origin: A&M §8.1 opening; H §18.3b; S §5.1.3

Theorem 9.14 (Order algebra). The relation $\succeq$ is reflexive, antisymmetric, and transitive, hence a partial order on symmetric matrices. If $\mathbf{A} \succeq \mathbf{B}$, then, for every conformable $\mathbf{P}$,

$$\mathbf{P}'\mathbf{A}\mathbf{P} \succeq \mathbf{P}'\mathbf{B}\mathbf{P}.$$

If also $\mathbf{C} \succeq \mathbf{D}$ and $\alpha \geq 0$, then

$$\mathbf{A} + \mathbf{C} \succeq \mathbf{B} + \mathbf{D}, \quad \alpha\mathbf{A} \succeq \alpha\mathbf{B}.$$

Origin: A&M §8.1 opening; H Lemmas 14.2.2–14.2.4 and Thm. 14.2.9

Proof. Reflexivity follows from $\mathbf{A} - \mathbf{A} = \mathbf{O} \succeq \mathbf{O}$. If $\mathbf{A} \succeq \mathbf{B}$ and $\mathbf{B} \succeq \mathbf{A}$, then

$$0 \leq \mathbf{x}'(\mathbf{A} - \mathbf{B})\mathbf{x} \leq 0 \quad (\mathbf{x} \in \mathbb{R}^n),$$

so $\mathbf{A} - \mathbf{B} = \mathbf{O}$ by Lemma 9.2. If $\mathbf{A} \succeq \mathbf{B} \succeq \mathbf{C}$, then

$$\mathbf{A} - \mathbf{C} = (\mathbf{A} - \mathbf{B}) + (\mathbf{B} - \mathbf{C}) \succeq \mathbf{O}.$$

For congruence,

$$\mathbf{x}'\mathbf{P}'(\mathbf{A} - \mathbf{B})\mathbf{P}\mathbf{x} = (\mathbf{P}\mathbf{x})'(\mathbf{A} - \mathbf{B})(\mathbf{P}\mathbf{x}) \geq 0.$$

For addition and nonnegative scaling,

$$\mathbf{x}'\{(\mathbf{A} + \mathbf{C}) - (\mathbf{B} + \mathbf{D})\}\mathbf{x} = \mathbf{x}'(\mathbf{A} - \mathbf{B})\mathbf{x} + \mathbf{x}'(\mathbf{C} - \mathbf{D})\mathbf{x} \geq 0,$$

$$\mathbf{x}'\{\alpha\mathbf{A} - \alpha\mathbf{B}\}\mathbf{x} = \alpha\mathbf{x}'(\mathbf{A} - \mathbf{B})\mathbf{x} \geq 0.$$

□

Theorem 9.15 (Spectral, rank, trace, and determinant consequences). Let $\mathbf{A} \succeq \mathbf{B}$, and arrange their eigenvalues as $\lambda_1 \geq \cdots \geq \lambda_n$. Then

$$\lambda_k(\mathbf{A}) \geq \lambda_k(\mathbf{B}), \quad k = 1, \dots, n,$$

and $\text{tr}(\mathbf{A}) \geq \text{tr}(\mathbf{B})$. If $\mathbf{A} \succeq \mathbf{B} \succeq \mathbf{O}$, then

$$\det(\mathbf{A}) \geq \det(\mathbf{B}), \quad \ker(\mathbf{A}) \subseteq \ker(\mathbf{B}), \quad \text{rank}(\mathbf{A}) \geq \text{rank}(\mathbf{B}).$$

Origin: A&M Ex. 12.8; H Cor. 18.1.8; S Thm. 6.3

Proof. The Courant–Fischer formula gives

$$\lambda_k(\mathbf{A}) = \min_{\dim \mathcal{S} = n-k+1} \max_{\mathbf{0} \neq \mathbf{x} \in \mathcal{S}} \frac{\mathbf{x}' \mathbf{A} \mathbf{x}}{\mathbf{x}' \mathbf{x}} \geq \min_{\dim \mathcal{S} = n-k+1} \max_{\mathbf{0} \neq \mathbf{x} \in \mathcal{S}} \frac{\mathbf{x}' \mathbf{B} \mathbf{x}}{\mathbf{x}' \mathbf{x}} = \lambda_k(\mathbf{B}).$$

Summing gives the trace inequality. Under semidefiniteness, every eigenvalue is nonnegative, so multiplying the componentwise inequalities gives the determinant inequality. If $\mathbf{x} \in \ker(\mathbf{A})$, then

$$0 \leq \mathbf{x}' \mathbf{B} \mathbf{x} \leq \mathbf{x}' \mathbf{A} \mathbf{x} = 0.$$

The null criterion in Theorem 9.4 gives $\mathbf{B}\mathbf{x} = \mathbf{0}$. Nullity and rank then give the last inequality. $\square$

Theorem 9.16 (Inverse reverses order). If $\mathbf{A} \succeq \mathbf{B} \succ \mathbf{0}$, then

$$\mathbf{B}^{-1} \succeq \mathbf{A}^{-1}.$$

Equality holds exactly when $\mathbf{A} = \mathbf{B}$; if $\mathbf{A} \succ \mathbf{B}$, then $\mathbf{B}^{-1} \succ \mathbf{A}^{-1}$.

Origin: A&M Ex. 12.16; H Thm. 18.3.4

Proof. Set $\mathbf{C} = \mathbf{B}^{-1/2}\mathbf{A}\mathbf{B}^{-1/2}$. Then $\mathbf{C} \succeq \mathbf{I}_n$, so the eigenvalues c_i of $\mathbf{C}$ satisfy $c_i \geq 1$. Hence

$$\mathbf{I}_n - \mathbf{C}^{-1} \preceq \mathbf{0}, \quad \mathbf{B}^{-1} - \mathbf{A}^{-1} = \mathbf{B}^{-1/2}(\mathbf{I}_n - \mathbf{C}^{-1})\mathbf{B}^{-1/2} \preceq \mathbf{0}.$$

The middle matrix is zero exactly when every $c_i = 1$, equivalently $\mathbf{C} = \mathbf{I}_n$ and $\mathbf{A} = \mathbf{B}$. If $\mathbf{A} - \mathbf{B} \succ \mathbf{0}$, then $\mathbf{C} - \mathbf{I}_n \succ \mathbf{0}$, every $c_i > 1$, and $\mathbf{I}_n - \mathbf{C}^{-1} \succ \mathbf{0}$. $\square$

2.3 Generalized eigenvalues

Definition 9.17 (Symmetric generalized eigenproblem). Let $\mathbf{A} = \mathbf{A}'$ and $\mathbf{B} = \mathbf{B}' \succ \mathbf{O}$. A scalar $\lambda \in \mathbb{R}$ and a vector $\mathbf{0} \neq \mathbf{x} \in \mathbb{R}^n$ form a **generalized eigenpair** of $(\mathbf{A}, \mathbf{B})$ if

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{B}\mathbf{x}.$$

Equivalently, $\det(\mathbf{A} - \lambda\mathbf{B}) = 0$.

Origin: H §21.14, Eqs. 21.14.3 and 21.14.6; A&M Ex. 8.37

Theorem 9.18 (Symmetric reduction and generalized Rayleigh quotient). Let

$$\mathbf{C} = \mathbf{B}^{-1/2}\mathbf{A}\mathbf{B}^{-1/2} = \mathbf{C}'.$$

The generalized eigenvalues of $(\mathbf{A}, \mathbf{B})$ are the real eigenvalues $\lambda_1 \geq \dots \geq \lambda_n$ of $\mathbf{C}$. There is a basis $\mathbf{x}_1, \dots, \mathbf{x}_n$ such that

$$\mathbf{A}\mathbf{x}_i = \lambda_i\mathbf{B}\mathbf{x}_i, \quad \mathbf{x}_i'\mathbf{B}\mathbf{x}_j = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Moreover,

$$\lambda_n = \min_{\mathbf{x} \neq \mathbf{0}} \frac{\mathbf{x}' \mathbf{A} \mathbf{x}}{\mathbf{x}' \mathbf{B} \mathbf{x}}, \quad \lambda_1 = \max_{\mathbf{x} \neq \mathbf{0}} \frac{\mathbf{x}' \mathbf{A} \mathbf{x}}{\mathbf{x}' \mathbf{B} \mathbf{x}},$$

and, for every $t \in \mathbb{R}$,

$$\mathbf{A} \succeq t\mathbf{B} \iff \lambda_n \geq t.$$

Finally,

$$\det(\mathbf{A} - t\mathbf{B}) = \det(\mathbf{B}) \det(\mathbf{C} - t\mathbf{I}_n).$$

Origin: A&M Ex. 8.37; H §21.14, Eqs. 21.14.1–21.14.7; S Prop. 6.1

Proof. Choose an orthonormal eigenbasis $\mathbf{u}_i$ of $\mathbf{C}$, and set $\mathbf{x}_i = \mathbf{B}^{-1/2} \mathbf{u}_i$. Multiplication of $\mathbf{C} \mathbf{u}_i = \lambda_i \mathbf{u}_i$ by $\mathbf{B}^{1/2}$ gives

$$\mathbf{A} \mathbf{x}_i = \lambda_i \mathbf{B} \mathbf{x}_i, \quad \mathbf{x}_i' \mathbf{B} \mathbf{x}_j = \mathbf{u}_i' \mathbf{u}_j.$$

Since $\mathbf{B}^{-1/2}$ is nonsingular, the $\mathbf{x}_i$ form a basis. With $\mathbf{y} = \mathbf{B}^{1/2} \mathbf{x}$,

$$\frac{\mathbf{x}' \mathbf{A} \mathbf{x}}{\mathbf{x}' \mathbf{B} \mathbf{x}} = \frac{\mathbf{y}' \mathbf{C} \mathbf{y}}{\mathbf{y}' \mathbf{y}};$$

the two extrema are the Rayleigh extrema of $\mathbf{C}$. Also

$$\mathbf{B}^{-1/2}(\mathbf{A} - t\mathbf{B})\mathbf{B}^{-1/2} = \mathbf{C} - t\mathbf{I}_n,$$

so congruence and the spectral characterisation give the order equivalence. The factorisation

$$\mathbf{A} - t\mathbf{B} = \mathbf{B}^{1/2}(\mathbf{C} - t\mathbf{I}_n)\mathbf{B}^{1/2}$$

gives the determinant identity. □

3 Block covariance structure

3.1 Block definiteness

Theorem 9.19 (Schur-complement criterion). Let

$$\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}' & \mathbf{C} \end{pmatrix}, \quad \mathbf{A} = \mathbf{A}' \succ \mathbf{O}, \quad \mathbf{C} = \mathbf{C}',$$

and set $\mathbf{S} = \mathbf{C} - \mathbf{B}'\mathbf{A}^{-1}\mathbf{B}$. Then

$$\mathbf{M} \succeq \mathbf{O} \iff \mathbf{S} \succeq \mathbf{O}, \quad \mathbf{M} \succ \mathbf{O} \iff \mathbf{S} \succ \mathbf{O}.$$

More generally, for every symmetric $\mathbf{W}$ conformable with $\mathbf{C}$,

$$\mathbf{M} - \begin{pmatrix} \mathbf{O} & \mathbf{O} \\ \mathbf{O} & \mathbf{W} \end{pmatrix} \succeq \mathbf{O} \iff \mathbf{S} \succeq \mathbf{W}.$$

Origin: H Cor. 14.8.6; S Ex. 11.6

Proof. The block factorisation is

$$\mathbf{M} = \underbrace{\begin{pmatrix} \mathbf{I} & \mathbf{O} \\ \mathbf{B}'\mathbf{A}^{-1} & \mathbf{I} \end{pmatrix}}_{\mathbf{P}} \begin{pmatrix} \mathbf{A} & \mathbf{O} \\ \mathbf{O} & \mathbf{S} \end{pmatrix} \underbrace{\begin{pmatrix} \mathbf{I} & \mathbf{A}^{-1}\mathbf{B} \\ \mathbf{O} & \mathbf{I} \end{pmatrix}}_{\mathbf{P}'}.$$

The matrix $\mathbf{P}$ is nonsingular. Congruence preserves definiteness, and

$$\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix}' \begin{pmatrix} \mathbf{A} & \mathbf{O} \\ \mathbf{O} & \mathbf{S} \end{pmatrix} \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} = \mathbf{x}'\mathbf{A}\mathbf{x} + \mathbf{y}'\mathbf{S}\mathbf{y}.$$

This proves both equivalences. Replacing $\mathbf{C}$ by $\mathbf{C} - \mathbf{W}$ replaces $\mathbf{S}$ by $\mathbf{S} - \mathbf{W}$, proving the last assertion. $\square$

3.2 Expectation and covariance transformations

Definition 9.20 (Expectation and covariance). On a real vector space $\mathcal{L}$ of real random variables, an **expectation** is a positive linear map $\mathbf{E} : \mathcal{L} \rightarrow \mathbb{R}$ with $\mathbf{E}[1] = 1$: whenever the displayed variables belong to $\mathcal{L}$,

$$\mathbf{E}[aU + bV] = a \mathbf{E}[U] + b \mathbf{E}[V], \quad U \geq 0 \implies \mathbf{E}[U] \geq 0.$$

For random vectors $\mathbf{X} \in \mathbb{R}^p$ and $\mathbf{Y} \in \mathbb{R}^q$, with all displayed expectations finite, define

$$\boldsymbol{\mu}_{\mathbf{X}} = \mathbf{E}[\mathbf{X}] := (\mathbf{E}[X_1], \dots, \mathbf{E}[X_p])', \quad \boldsymbol{\mu}_{\mathbf{Y}} = \mathbf{E}[\mathbf{Y}] := (\mathbf{E}[Y_1], \dots, \mathbf{E}[Y_q])',$$

$$\text{cov}(\mathbf{X}, \mathbf{Y}) = \mathbf{E}[(\mathbf{X} - \boldsymbol{\mu}_{\mathbf{X}})(\mathbf{Y} - \boldsymbol{\mu}_{\mathbf{Y}})'], \quad \text{cov}(\mathbf{X}) = \text{cov}(\mathbf{X}, \mathbf{X}).$$

The first matrix is the **cross-covariance**; the second is the **covariance matrix**. For a scalar U , $\text{var}(U) = \mathbf{E}[(U - \mathbf{E}[U])^2]$. For a positive integer r and $\mathbf{W} \in \mathbb{R}^{r \times p}$, set

$$\mathbf{Z} = \mathbf{W}(\mathbf{X} - \boldsymbol{\mu}_{\mathbf{X}}).$$

If $\text{cov}(\mathbf{Z}) = \mathbf{I}_r$, then $\mathbf{W}$ is a **whitening matrix** for $\mathbf{X}$, and $\mathbf{Z}$ is the **whitened vector**.

Origin: Course formulation

Theorem 9.21 (Covariance calculus). Let $\mathbf{X} \in \mathbb{R}^p$, $\mathbf{Y} \in \mathbb{R}^q$, and let all displayed second moments be finite. For constant matrices $\mathbf{C}, \mathbf{D}$ and constant vectors $\mathbf{c}, \mathbf{d}$,

$$\mathbf{E}[\mathbf{C}\mathbf{X} + \mathbf{c}] = \mathbf{C}\boldsymbol{\mu}_{\mathbf{X}} + \mathbf{c},$$

$$\text{cov}(\mathbf{C}\mathbf{X} + \mathbf{c}, \mathbf{D}\mathbf{Y} + \mathbf{d}) = \mathbf{C} \text{cov}(\mathbf{X}, \mathbf{Y}) \mathbf{D}',$$

$$\text{cov}(\mathbf{Y}, \mathbf{X}) = \text{cov}(\mathbf{X}, \mathbf{Y})'.$$

The matrix $\Sigma = \text{cov}(\mathbf{X})$ is symmetric positive semidefinite, and

$$\mathbf{a}'\Sigma\mathbf{b} = \text{cov}(\mathbf{a}'\mathbf{X}, \mathbf{b}'\mathbf{X}), \quad \ker(\Sigma) = \{\mathbf{a} : \text{var}(\mathbf{a}'\mathbf{X}) = 0\}.$$

Origin: Course formulation from H §14.2a

Proof. Linearity of expectation gives the first identity. Subtracting that mean gives

$$(\mathbf{C}\mathbf{X} + \mathbf{c}) - \mathbf{E}[\mathbf{C}\mathbf{X} + \mathbf{c}] = \mathbf{C}(\mathbf{X} - \boldsymbol{\mu}_{\mathbf{X}}),$$

and similarly for $\mathbf{Y}$; substitution gives the second identity. Transposition gives the third. With $\tilde{\mathbf{X}} = \mathbf{X} - \boldsymbol{\mu}_{\mathbf{X}}$,

$$\Sigma' = \mathbf{E}[(\tilde{\mathbf{X}}\tilde{\mathbf{X}}')'] = \Sigma, \quad \mathbf{a}'\Sigma\mathbf{a} = \mathbf{E}[(\mathbf{a}'\tilde{\mathbf{X}})^2] \geq 0.$$

Moreover,

$$\mathbf{a}'\Sigma\mathbf{b} = \mathbf{E}[(\mathbf{a}'\tilde{\mathbf{X}})(\mathbf{b}'\tilde{\mathbf{X}})] = \text{cov}(\mathbf{a}'\mathbf{X}, \mathbf{b}'\mathbf{X}).$$

Finally,

$$\text{var}(\mathbf{a}'\mathbf{X}) = \mathbf{a}'\Sigma\mathbf{a} = 0 \iff \Sigma\mathbf{a} = \mathbf{0}$$

by the null criterion for a positive semidefinite matrix. □

Theorem 9.22 (Whitening, including singular covariance). Let $\mathbf{X} \in \mathbb{R}^p$, let $\text{cov}(\mathbf{X}) = \mathbf{\Sigma} \succeq \mathbf{0}$, and set $r = \text{rank}(\mathbf{\Sigma})$, so $0 \leq r \leq p$. If $r = 0$, then $\mathbf{\Sigma} = \mathbf{0}$, and there are no nonzero-variance coordinates to whiten. If $r > 0$, write the positive spectral part as

$$\mathbf{\Sigma} = \mathbf{Q}_r \mathbf{\Lambda}_r \mathbf{Q}_r', \quad \mathbf{Q}_r' \mathbf{Q}_r = \mathbf{I}_r, \quad \mathbf{\Lambda}_r = \text{diag}(\lambda_1, \dots, \lambda_r) \succ \mathbf{0}.$$

Then

$$\mathbf{W}_r = \mathbf{\Lambda}_r^{-1/2} \mathbf{Q}_r', \quad \mathbf{Z} = \mathbf{W}_r (\mathbf{X} - \boldsymbol{\mu}_{\mathbf{X}})$$

satisfies

$$\mathbb{E}[\mathbf{Z}] = \mathbf{0}, \quad \text{cov}(\mathbf{Z}) = \mathbf{I}_r.$$

If $r = p$, the square whitening matrix $\mathbf{\Sigma}^{-1/2}$ gives the whitened vector

$$\tilde{\mathbf{Z}} = \mathbf{\Sigma}^{-1/2} (\mathbf{X} - \boldsymbol{\mu}_{\mathbf{X}}) = \mathbf{Q}_r \mathbf{Z}, \quad \text{cov}(\tilde{\mathbf{Z}}) = \mathbf{I}_p.$$

Conversely, let s be a positive integer. If $\mathbf{Z} \in \mathbb{R}^s$, $\mathbb{E}[\mathbf{Z}] = \mathbf{0}$, $\text{cov}(\mathbf{Z}) = \mathbf{I}_s$, $\mathbf{C} \in \mathbb{R}^{p \times s}$, and $\boldsymbol{\mu} \in \mathbb{R}^p$, then

$$\mathbf{Y} := \boldsymbol{\mu} + \mathbf{CZ} \implies \mathbb{E}[\mathbf{Y}] = \boldsymbol{\mu}, \quad \text{cov}(\mathbf{Y}) = \mathbf{C}\mathbf{C}'.$$

Origin: Course formulation from A&M Exs. 8.21 and 8.24; H Thm. 21.9.1

Proof. If $r = 0$, the spectral characterisation gives $\Sigma = \mathbf{O}$, and

$$\text{var}(\mathbf{a}'\mathbf{X}) = \mathbf{a}'\Sigma\mathbf{a} = 0 \quad (\mathbf{a} \in \mathbb{R}^p).$$

Let $r > 0$. Covariance calculus gives

$$\text{cov}(\mathbf{Z}) = \Lambda_r^{-1/2} \mathbf{Q}_r' \Sigma \mathbf{Q}_r \Lambda_r^{-1/2} = \Lambda_r^{-1/2} \Lambda_r \Lambda_r^{-1/2} = \mathbf{I}_r,$$

and

$$\mathbf{E}[\mathbf{Z}] = \mathbf{W}_r \{\mathbf{E}[\mathbf{X}] - \boldsymbol{\mu}_X\} = \mathbf{0}.$$

If $r = p$, functional calculus gives $\mathbf{Q}_r \Lambda_r^{-1/2} \mathbf{Q}_r' = \Sigma^{-1/2}$, hence $\tilde{\mathbf{Z}} = \mathbf{Q}_r \mathbf{Z}$; orthogonal congruence preserves $\mathbf{I}_p$. The converse is

$$\mathbf{E}[\mathbf{Y}] = \mathbf{E}[\boldsymbol{\mu} + \mathbf{CZ}] = \boldsymbol{\mu}, \quad \text{cov}(\mathbf{Y}) = \text{cov}(\boldsymbol{\mu} + \mathbf{CZ}) = \mathbf{C} \mathbf{I}_s \mathbf{C}' = \mathbf{C} \mathbf{C}'.$$

□

3.3 Residual covariance and Mahalanobis distance

Theorem 9.23 (Minimum linear-residual covariance). Let the joint covariance of $\mathbf{X} \in \mathbb{R}^p$ and $\mathbf{Y} \in \mathbb{R}^q$ be

$$\text{cov} \begin{pmatrix} \mathbf{X} \\ \mathbf{Y} \end{pmatrix} = \begin{pmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_{YY} \end{pmatrix}, \quad \Sigma_{XX} \succ \mathbf{0}.$$

For $\mathbf{B} \in \mathbb{R}^{q \times p}$, set

$$\mathbf{R}_\mathbf{B} = (\mathbf{Y} - \mu_\mathbf{Y}) - \mathbf{B}(\mathbf{X} - \mu_\mathbf{X}), \quad \mathbf{B}_* = \Sigma_{YX} \Sigma_{XX}^{-1},$$

and

$$\mathbf{S} = \Sigma_{YY} - \Sigma_{YX} \Sigma_{XX}^{-1} \Sigma_{XY}.$$

Then $\mathbf{S} \succeq \mathbf{0}$ and

$$\text{cov}(\mathbf{R}_\mathbf{B}) = \mathbf{S} + (\mathbf{B} - \mathbf{B}_*) \Sigma_{XX} (\mathbf{B} - \mathbf{B}_*)' \succeq \mathbf{S}.$$

Equality holds exactly when $\mathbf{B} = \mathbf{B}_*$.

Origin: Course formulation from H Cor. 14.8.6; S Ex. 11.6

Proof. The joint covariance is positive semidefinite by covariance calculus, so the block criterion gives $\mathbf{S} \succeq \mathbf{O}$. Direct expansion gives

$$\text{cov}(\mathbf{R}_\mathbf{B}) = \Sigma_{YY} - \mathbf{B}\Sigma_{XY} - \Sigma_{YX}\mathbf{B}' + \mathbf{B}\Sigma_{XX}\mathbf{B}'.$$

Since $\mathbf{B}_*\Sigma_{XX} = \Sigma_{YX}$ and $\Sigma_{XX}\mathbf{B}'_* = \Sigma_{XY}$, completing the matrix square gives the stated identity. The added term is positive semidefinite. It is the Gram matrix of $\Sigma_{XX}^{1/2}(\mathbf{B} - \mathbf{B}_*)'$, hence is zero if and only if

$$\Sigma_{XX}^{1/2}(\mathbf{B} - \mathbf{B}_*)' = \mathbf{O},$$

equivalently $\mathbf{B} = \mathbf{B}_*$, because $\Sigma_{XX}^{1/2}$ is nonsingular. □

Definition 9.24 (Mahalanobis distance). For $\Sigma = \Sigma' \succ \mathbf{O}$ and $\mathbf{x}, \boldsymbol{\mu} \in \mathbb{R}^p$, define

$$d_\Sigma(\mathbf{x}, \boldsymbol{\mu}) = \sqrt{(\mathbf{x} - \boldsymbol{\mu})'\Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})}.$$

Equivalently,

$$d_{\Sigma}(\mathbf{x}, \boldsymbol{\mu}) = \|\Sigma^{-1/2}(\mathbf{x} - \boldsymbol{\mu})\|.$$

Origin: Course formulation from H §14.10a and Thm. 21.9.1

Theorem 9.25 (Affine invariance of Mahalanobis distance). Let $\mathbf{C} \in \mathbb{R}^{p \times p}$ be nonsingular, $\mathbf{d} \in \mathbb{R}^p$, and

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{d}, \quad \boldsymbol{\nu} = \mathbf{C}\boldsymbol{\mu} + \mathbf{d}, \quad \boldsymbol{\Omega} = \mathbf{C}\Sigma\mathbf{C}'.$$

Then $\boldsymbol{\Omega} \succ \mathbf{O}$ and

$$d_{\boldsymbol{\Omega}}(\mathbf{y}, \boldsymbol{\nu}) = d_{\Sigma}(\mathbf{x}, \boldsymbol{\mu}).$$

Origin: Course formulation from A&M Ex. 8.28; H Cor. 14.2.10

Proof. Congruence gives $\boldsymbol{\Omega} \succ \mathbf{O}$, and

$$\boldsymbol{\Omega}^{-1} = (\mathbf{C}')^{-1}\Sigma^{-1}\mathbf{C}^{-1}.$$

Since $\mathbf{y} - \boldsymbol{\nu} = \mathbf{C}(\mathbf{x} - \boldsymbol{\mu})$,

$$(\mathbf{y} - \boldsymbol{\nu})'\boldsymbol{\Omega}^{-1}(\mathbf{y} - \boldsymbol{\nu}) = (\mathbf{x} - \boldsymbol{\mu})'\Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu}).$$

Both distances are the nonnegative square roots of these quantities. □

Example 9.3 (A scalar linear residual). Suppose

$$\text{cov} \begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix}.$$

For $R_b = (Y - \mathbf{E}[Y]) - b(X - \mathbf{E}[X])$,

$$b_* = \frac{2}{4} = \frac{1}{2}, \quad \text{var}(R_b) = 3 - 4b + 4b^2 = 2 + 4 \left(b - \frac{1}{2}\right)^2.$$

Thus the minimum residual variance is the Schur complement $3 - 2(4^{-1})2 = 2$.

Origin: Course formulation from H Cor. 14.8.6

4 Computation: Numerical Checks in R

4.1 Eigenvalue, minor, and Cholesky diagnostics

```
1 A_pd <- rbind(c(4, 2, 0), c(2, 5, 1), c(0, 1, 2))
2 A_psd <- rbind(c(1, 1, 0), c(1, 2, 0), c(0, 0, 0))
```

```

3 A_ind <- rbind(c(0, 0, 1), c(0, -1, 0), c(1, 0, 0))
4
5 diagnose <- function(A, tol = 1e-10) {
6   ev <- eigen(A, symmetric = TRUE)$values
7   leading <- vapply(seq_len(nrow(A)), function(k)
8     det(A[seq_len(k), seq_len(k), drop = FALSE]), numeric(1))
9   c(min_eigenvalue = min(ev),
10     min_leading_minor = min(leading),
11     numerical_rank = qr(A, tol = tol)$rank,
12     chol_success = as.numeric(!inherits(try(chol(A), silent = TRUE),
13                                           "try-error")))
14 }
15 rbind(positive_definite = diagnose(A_pd),
16       positive_semidefinite = diagnose(A_psd),
17       indefinite = diagnose(A_ind))

```

```

##                min_eigenvalue min_leading_minor
## positive_definite           1.482                4
## positive_semidefinite         0.000                0
## indefinite                  -1.000                0
##                numerical_rank chol_success

```

## positive_definite	3	1
## positive_seminefinite	2	0
## indefinite	3	0

The numerical rank uses the QR method in `qr()` with tolerance 10^{-10} . Eigenvalues in $[-10^{-10}, 0)$ are treated as roundoff for a numerical semidefinite decision. Both $\mathbf{A}_{\text{psd}}$ and $\mathbf{A}_{\text{ind}}$ have minimum leading minor zero; the negative eigenvalue of $\mathbf{A}_{\text{ind}}$ separates them and confirms that nonnegative leading minors do not test semidefiniteness.

4.2 Cholesky factor and symmetric square root

```

1 A <- rbind(c(2, 1), c(1, 2))
2 L <- t(chol(A))
3 eA <- eigen(A, symmetric = TRUE)
4 R <- eA$vectors %*% diag(sqrt(eA$values)) %*% t(eA$vectors)
5 c(cholesky_reconstruction = max(abs(L %*% t(L) - A)),
6   root_reconstruction = max(abs(R %*% R - A)),
7   factor_difference = max(abs(L - R)))

```

## cholesky_reconstruction	root_reconstruction
## 4.441e-16	6.661e-16

```
##          factor_difference
##          3.660e-01

1 A0 <- rbind(c(1, 1, 0), c(1, 2, 0), c(0, 0, 0))
2 L0 <- rbind(c(1, 0, 0), c(1, 1, 0), c(0, 0, 0))
3 e0 <- eigen(A0, symmetric = TRUE)
4 R0 <- e0$vectors %*% diag(sqrt(pmax(e0$values, 0))) %*% t(e0$vectors)
5 c(semidefinite_cholesky_residual = max(abs(L0 %*% t(L0) - A0)),
6   semidefinite_root_residual = max(abs(R0 %*% R0 - A0)),
7   base_chol_success = as.numeric(!inherits(try(chol(A0), silent = TRUE),
8                                             "try-error"))))

## semidefinite_cholesky_residual      semidefinite_root_residual
##                                0.000e+00                    6.661e-16
##          base_chol_success
##          0.000e+00
```

The positive part `pmax(e0$values, 0)` removes only roundoff-sized negative eigenvalues introduced by floating-point eigensolution.

4.3 Order reversal and generalized eigenvalues

```

1 B <- rbind(c(2, 0.5), c(0.5, 1.5))
2 u <- c(1, -2)
3 A <- B + tcrossprod(u)
4 eB <- eigen(B, symmetric = TRUE)
5 Binvhalf <- eB$vectors %*% diag(eB$values^(-1 / 2)) %*% t(eB$vectors)
6 C <- Binvhalf %*% A %*% Binvhalf
7 lambda_symmetric <- sort(eigen(C, symmetric = TRUE)$values,
8                           decreasing = TRUE)
9 lambda_direct <- sort(Re(eigen(solve(B, A))$values), decreasing = TRUE)
10 c(min_eigen_A_minus_B = min(eigen(A - B, symmetric = TRUE)$values),
11    min_eigen_inverse_gap = min(eigen(solve(B) - solve(A),
12                                     symmetric = TRUE)$values),
13    generalized_route_difference = max(abs(lambda_symmetric - lambda_direct)))

```

```

##          min_eigen_A_minus_B          min_eigen_inverse_gap
##                0.000e+00                -1.388e-17
## generalized_route_difference
##                3.553e-15

```

```
1 lambda_symmetric
```

```
## [1] 5.182 1.000
```

The generalized problem is reduced to the symmetric matrix $\mathbf{B}^{-1/2}\mathbf{A}\mathbf{B}^{-1/2}$; `solve(B, A)` is printed only as an additional eigenvalue check. Order-test eigenvalues no smaller than -10^{-12} are treated as zero.

4.4 Schur complement and residual covariance

```
1 Sigma_XX <- rbind(c(2, 0.5), c(0.5, 1))
2 Sigma_YX <- matrix(c(0.6, 0.4), nrow = 1)
3 Sigma_YY <- matrix(2, 1, 1)
4 Sigma_joint <- rbind(cbind(Sigma_XX, t(Sigma_YX)),
5                      cbind(Sigma_YX, Sigma_YY))
6 B_star <- Sigma_YX %*% solve(Sigma_XX)
7 S <- Sigma_YY - B_star %*% t(Sigma_YX)
8 B_trial <- matrix(c(1, -0.5), nrow = 1)
9 residual_direct <- Sigma_YY - B_trial %*% t(Sigma_YX) -
10  Sigma_YX %*% t(B_trial) + B_trial %*% Sigma_XX %*% t(B_trial)
```

```

11 residual_completed <- S +
12   (B_trial - B_star) %*% Sigma_XX %*% t(B_trial - B_star)
13 c(min_joint_eigenvalue = min(eigen(Sigma_joint, symmetric = TRUE)$values),
14   schur_complement = S,
15   completion_difference = max(abs(residual_direct - residual_completed)),
16   excess_residual_variance = residual_direct - S)

```

```

##      min_joint_eigenvalue      schur_complement
##              0.7707              1.7486
##      completion_difference excess_residual_variance
##              0.0000              1.2014

```

4.5 Deterministic whitening and Mahalanobis distance

```

1 Sigma <- rbind(c(4, 1.2), c(1.2, 2))
2 mu <- c(3, -1)
3 eS <- eigen(Sigma, symmetric = TRUE)
4 Sroot <- eS$vectors %*% diag(sqrt(eS$values)) %*% t(eS$vectors)
5 Sinvroot <- eS$vectors %*% diag(eS$values^(-1 / 2)) %*% t(eS$vectors)
6 a <- sqrt(3 / 2)
7 Z <- rbind(c(a, 0), c(-a, 0), c(0, a), c(0, -a))
8 X <- sweep(Z %*% Sroot, 2, mu, "+")

```

```

9 X_centered <- sweep(X, 2, mu, "-")
10 Z_white <- X_centered %*% Sinvroot
11 U <- chol(Sigma)
12 d2_chol <- colSums(forwardsolve(t(U), t(X_centered))^2)
13 d2_package <- mahalanobis(X, center = mu, cov = Sigma)
14 c(covariance_construction = max(abs(cov(X) - Sigma)),
15   whitening_residual = max(abs(cov(Z_white) - diag(2))),
16   mahalanobis_difference = max(abs(d2_chol - d2_package)))

```

```

## covariance_construction      whitening_residual
##                1.776e-15                8.882e-16
## mahalanobis_difference
##                2.220e-16

```

The four rows of $\mathbf{Z}$ have sample mean zero and sample covariance $\mathbf{I}_2$ exactly; sampling error is absent from both covariance comparisons.

5 Pitfalls

5.1 Nonnegative leading minors do not test semidefiniteness

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad (\Delta_1, \Delta_2, \Delta_3) = (0, 0, 1),$$

$$(\mathbf{e}_1 + \mathbf{e}_3)' \mathbf{A} (\mathbf{e}_1 + \mathbf{e}_3) = 2, \quad \mathbf{e}_2' \mathbf{A} \mathbf{e}_2 = -1.$$

Sylvester's criterion uses strict positivity for positive definiteness. A semidefinite criterion cannot be obtained by replacing $>$ with $\geq$ only for the leading minors.

5.2 The Löwner order is not entrywise

$$\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \succeq \mathbf{0}, \quad \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \text{ has eigenvalues } 3, -1.$$

A positive semidefinite matrix may have negative entries; an entrywise nonnegative symmetric matrix may be indefinite.

5.3 The order is partial and squaring need not preserve it

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

so the two diagonal matrices are incomparable. Also, with

$$\mathbf{B} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix},$$

$$\mathbf{A} - \mathbf{B} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \succeq \mathbf{0}, \quad \mathbf{A}^2 - \mathbf{B}^2 = \begin{pmatrix} 4 & 3 \\ 3 & 2 \end{pmatrix}, \quad \det(\mathbf{A}^2 - \mathbf{B}^2) = -1.$$

5.4 A singular covariance has no inverse-based distance

$$\mathbf{\Sigma} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \det(\mathbf{\Sigma}) = 0, \quad \text{rank}(\mathbf{\Sigma}) = 1.$$

The expression $(\mathbf{x} - \boldsymbol{\mu})' \mathbf{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})$ is undefined. The rank-reduced whitening matrix in Theorem 9.22 remains defined.

6 Exercises

Exercise 9.1 (Recovering a symmetric matrix from its quadratic form). Let $\mathbf{A} = (a_{ij}) = \mathbf{A}'$ and suppose that the values of $q_{\mathbf{A}}$ are known. Recover every entry of $\mathbf{A}$ from the values at $\mathbf{e}_i$ and $\mathbf{e}_i + \mathbf{e}_j$.

Origin: A&M Ex. 8.3; H Cor. 14.1.3

Solution.

$$a_{ii} = q_{\mathbf{A}}(\mathbf{e}_i),$$

and, for $i < j$,

$$q_{\mathbf{A}}(\mathbf{e}_i + \mathbf{e}_j) = a_{ii} + 2a_{ij} + a_{jj},$$

so

$$a_{ij} = a_{ji} = \frac{1}{2}\{q_{\mathbf{A}}(\mathbf{e}_i + \mathbf{e}_j) - q_{\mathbf{A}}(\mathbf{e}_i) - q_{\mathbf{A}}(\mathbf{e}_j)\}.$$

These $n + n(n-1)/2$ values determine all symmetric entries. 

Exercise 9.2 (Complete 2×2 semidefinite test). For

$$\mathbf{A} = \begin{pmatrix} a & b \\ b & c \end{pmatrix},$$

prove

$$\mathbf{A} \succeq \mathbf{0} \iff a \geq 0, \quad c \geq 0, \quad b^2 \leq ac.$$

State the corresponding positive definite condition.

Origin: A&M Ex. 8.7; H Cor. 14.5.14 and Thm. 14.9.5

Solution. Necessity follows from Lemma 9.6. Conversely, if $a > 0$,

$$ax^2 + 2bxy + cy^2 = a \left(x + \frac{b}{a}y \right)^2 + \left(c - \frac{b^2}{a} \right) y^2 \geq 0.$$

If $a = 0$, then $b^2 \leq ac = 0$, so $b = 0$ and the form is $cy^2 \geq 0$. Sylvester's criterion gives

$$\mathbf{A} \succ \mathbf{0} \iff a > 0 \text{ and } ac - b^2 > 0.$$



Exercise 9.3 (Kernel of a semidefinite sum). If $\mathbf{A}, \mathbf{B} \succeq \mathbf{O}$, prove

$$\ker(\mathbf{A} + \mathbf{B}) = \ker(\mathbf{A}) \cap \ker(\mathbf{B}).$$

Deduce that $\mathbf{A} + \mathbf{B} \succ \mathbf{O}$ exactly when $\ker(\mathbf{A}) \cap \ker(\mathbf{B}) = \{\mathbf{0}\}$.

Origin: A&M Ex. 8.8; H Lemma 14.2.4

Solution. If $(\mathbf{A} + \mathbf{B})\mathbf{x} = \mathbf{0}$, then

$$0 = \mathbf{x}'(\mathbf{A} + \mathbf{B})\mathbf{x} = \mathbf{x}'\mathbf{A}\mathbf{x} + \mathbf{x}'\mathbf{B}\mathbf{x}.$$

Both summands are nonnegative, hence both are zero. The null criterion gives $\mathbf{A}\mathbf{x} = \mathbf{B}\mathbf{x} = \mathbf{0}$. Conversely, these two equalities give $(\mathbf{A} + \mathbf{B})\mathbf{x} = \mathbf{0}$. Since $\mathbf{A} + \mathbf{B} \succeq \mathbf{O}$, it is positive definite exactly when its kernel is zero. 

Exercise 9.4 (Low-order principal minors are insufficient). Let $\mathbf{1} = (1, 1, 1)'$ and

$$\mathbf{A} = \frac{7}{4}\mathbf{I}_3 - \frac{3}{4}\mathbf{1}\mathbf{1}'.$$

Show that every principal minor of order one or two is positive, but that $\mathbf{A}$ is indefinite.

Origin: A&M Ex. 8.32; H Thm. 14.9.11

Solution. Every diagonal entry is 1, and every principal submatrix of order two is

$$\begin{pmatrix} 1 & -3/4 \\ -3/4 & 1 \end{pmatrix}, \quad \det \begin{pmatrix} 1 & -3/4 \\ -3/4 & 1 \end{pmatrix} = 1 - \frac{9}{16} = \frac{7}{16} > 0.$$

Since $\mathbf{1}\mathbf{1}' = 3\mathbf{1}$,

$$\mathbf{A}\mathbf{1} = -\frac{1}{2}\mathbf{1}.$$

For every $\mathbf{x} \perp \mathbf{1}$, $\mathbf{1}'\mathbf{x} = 0$ and $\mathbf{A}\mathbf{x} = (7/4)\mathbf{x}$. Hence the eigenvalues are $-1/2, 7/4, 7/4$, so $\mathbf{A}$ is indefinite. ■

Exercise 9.5 (Rank-one positive update). Let $\mathbf{A} \succ \mathbf{O}$ and $\mathbf{u} \in \mathbb{R}^n$. Prove

$$\mathbf{A} + \mathbf{u}\mathbf{u}' \succ \mathbf{O}, \quad \det(\mathbf{A} + \mathbf{u}\mathbf{u}') = \det(\mathbf{A})\{1 + \mathbf{u}'\mathbf{A}^{-1}\mathbf{u}\}.$$

When is the determinant unchanged?

Origin: A&M Ex. 8.35; H Thm. 18.1.6

Solution. For $\mathbf{x} \neq \mathbf{0}$,

$$\mathbf{x}'(\mathbf{A} + \mathbf{u}\mathbf{u}')\mathbf{x} = \mathbf{x}'\mathbf{A}\mathbf{x} + (\mathbf{u}'\mathbf{x})^2 > 0.$$

The rank-one determinant identity from partitioned matrices gives

$$\det(\mathbf{A} + \mathbf{u}\mathbf{u}') = \det(\mathbf{A})(1 + \mathbf{u}'\mathbf{A}^{-1}\mathbf{u}).$$

Since $\mathbf{A}^{-1} \succ \mathbf{0}$, $\mathbf{u}'\mathbf{A}^{-1}\mathbf{u} = 0 \iff \mathbf{u} = \mathbf{0}$. This is exactly the equality case. ■

Exercise 9.6 (Null space of the square root). If $\mathbf{A} \succeq \mathbf{0}$, prove

$$\ker(\mathbf{A}^{1/2}) = \ker(\mathbf{A}), \quad \text{col}(\mathbf{A}^{1/2}) = \text{col}(\mathbf{A}), \quad \text{rank}(\mathbf{A}^{1/2}) = \text{rank}(\mathbf{A}).$$

Origin: A&M Ex. 8.24; H Thm. 21.9.1

Solution. For an orthogonal $\mathbf{Q}$,

$$\mathbf{Q}'\mathbf{A}\mathbf{Q} = \text{diag}(\lambda_1, \dots, \lambda_n), \quad \mathbf{Q}'\mathbf{A}^{1/2}\mathbf{Q} = \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n}).$$

The two diagonal entries vanish for the same indices. Hence the two kernels are equal and both

ranks count the same nonzero indices. For symmetric matrices, $\text{col}(\mathbf{A}) = \ker(\mathbf{A})^\perp$; equal kernels give equal column spaces. 

Exercise 9.7 (Inverse order equivalence). For $\mathbf{A}, \mathbf{B} \succ \mathbf{O}$, prove

$$\mathbf{A} \succeq \mathbf{B} \iff \mathbf{B}^{-1} \succeq \mathbf{A}^{-1}, \quad \mathbf{A} \succ \mathbf{B} \iff \mathbf{B}^{-1} \succ \mathbf{A}^{-1}.$$

Origin: A&M Ex. 12.16; H Thm. 18.3.4

Solution. The forward implications are Theorem 9.16. Conversely, apply that theorem to

$$\mathbf{B}^{-1} \succeq \mathbf{A}^{-1} \succ \mathbf{O} \quad \text{or} \quad \mathbf{B}^{-1} \succ \mathbf{A}^{-1} \succ \mathbf{O}.$$

It gives

$$(\mathbf{A}^{-1})^{-1} \succeq (\mathbf{B}^{-1})^{-1} \quad \text{or} \quad (\mathbf{A}^{-1})^{-1} \succ (\mathbf{B}^{-1})^{-1},$$

which are the required reverse implications. 

Exercise 9.8 (All square whitening matrices). Let $\Sigma \succ \mathbf{O}$ and $\mathbf{W} \in \mathbb{R}^{p \times p}$. Prove

$$\mathbf{W}\Sigma\mathbf{W}' = \mathbf{I}_p \iff \mathbf{W} = \mathbf{Q}\Sigma^{-1/2} \text{ for some orthogonal } \mathbf{Q}.$$

Origin: Course formulation from H Thm. 21.9.1

Solution. If $\mathbf{W}\Sigma\mathbf{W}' = \mathbf{I}_p$, set $\mathbf{Q} = \mathbf{W}\Sigma^{1/2}$. Then

$$\mathbf{Q}\mathbf{Q}' = \mathbf{W}\Sigma\mathbf{W}' = \mathbf{I}_p.$$

Thus $\mathbf{Q}$ is square and orthogonal, and $\mathbf{W} = \mathbf{Q}\Sigma^{-1/2}$. Conversely, if $\mathbf{Q}\mathbf{Q}' = \mathbf{I}_p$, then

$$\mathbf{W}\Sigma\mathbf{W}' = \mathbf{Q}\Sigma^{-1/2}\Sigma\Sigma^{-1/2}\mathbf{Q}' = \mathbf{Q}\mathbf{Q}' = \mathbf{I}_p.$$



Exercise 9.9 (Generalized eigenvalue interval). Let $\mathbf{A} = \mathbf{A}'$, $\mathbf{B} \succ \mathbf{O}$, and $m \leq M$. Prove

$$m\mathbf{B} \preceq \mathbf{A} \preceq M\mathbf{B} \iff m \leq \lambda_i \leq M \quad (i = 1, \dots, n),$$

where λ_i are the generalized eigenvalues of $(\mathbf{A}, \mathbf{B})$.

Origin: H Exs. 21.31–21.32

Solution. With $\mathbf{C} = \mathbf{B}^{-1/2}\mathbf{A}\mathbf{B}^{-1/2}$, congruence gives

$$\mathbf{A} - m\mathbf{B} \succeq \mathbf{0} \iff \mathbf{C} - m\mathbf{I} \succeq \mathbf{0} \iff \lambda_i \geq m \quad (i = 1, \dots, n),$$

and

$$M\mathbf{B} - \mathbf{A} \succeq \mathbf{0} \iff M\mathbf{I} - \mathbf{C} \succeq \mathbf{0} \iff \lambda_i \leq M \quad (i = 1, \dots, n).$$

Combining the two equivalences proves the claim. 

Exercise 9.10 (A block norm criterion). Let $\mathbf{A} \succ \mathbf{0}$, $\mathbf{C} \succ \mathbf{0}$, and $\mathbf{B} \in \mathbb{R}^{p \times q}$. Set

$$\mathbf{K} = \mathbf{A}^{-1/2}\mathbf{B}\mathbf{C}^{-1/2}.$$

Prove

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}' & \mathbf{C} \end{pmatrix} \succeq \mathbf{0} \iff \|\mathbf{K}\mathbf{z}\| \leq \|\mathbf{z}\| \quad (\mathbf{z} \in \mathbb{R}^q).$$

Origin: H Cor. 14.8.6; S Ex. 11.6

Solution. The Schur complement of $\mathbf{A}$ is

$$\mathbf{C} - \mathbf{B}'\mathbf{A}^{-1}\mathbf{B} = \mathbf{C}^{1/2}(\mathbf{I}_q - \mathbf{K}'\mathbf{K})\mathbf{C}^{1/2}.$$

The block criterion and congruence give

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}' & \mathbf{C} \end{pmatrix} \succeq \mathbf{0} \iff \mathbf{I}_q - \mathbf{K}'\mathbf{K} \succeq \mathbf{0}.$$

For $\mathbf{z} \in \mathbb{R}^q$,

$$\mathbf{z}'(\mathbf{I}_q - \mathbf{K}'\mathbf{K})\mathbf{z} = \|\mathbf{z}\|^2 - \|\mathbf{K}\mathbf{z}\|^2,$$

which gives the stated equivalence. 

Exercise 9.11 (Covariance of a sum). For random vectors $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^p$, with all displayed second moments finite, prove

$$\text{cov}(\mathbf{X} + \mathbf{Y}) = \text{cov}(\mathbf{X}) + \text{cov}(\mathbf{Y}) + \text{cov}(\mathbf{X}, \mathbf{Y}) + \text{cov}(\mathbf{Y}, \mathbf{X}).$$

Give the corresponding identity for $\mathbf{X} - \mathbf{Y}$.

Origin: Course formulation

Solution. Set $\tilde{\mathbf{X}} = \mathbf{X} - \boldsymbol{\mu}_{\mathbf{X}}$ and $\tilde{\mathbf{Y}} = \mathbf{Y} - \boldsymbol{\mu}_{\mathbf{Y}}$. Then

$$\text{cov}(\mathbf{X} + \mathbf{Y}) = \mathbb{E}[(\tilde{\mathbf{X}} + \tilde{\mathbf{Y}})(\tilde{\mathbf{X}} + \tilde{\mathbf{Y}})'],$$

and expansion gives the four stated terms. Replacing $\tilde{\mathbf{Y}}$ by $-\tilde{\mathbf{Y}}$ gives

$$\text{cov}(\mathbf{X} - \mathbf{Y}) = \text{cov}(\mathbf{X}) + \text{cov}(\mathbf{Y}) - \text{cov}(\mathbf{X}, \mathbf{Y}) - \text{cov}(\mathbf{Y}, \mathbf{X}).$$



Exercise 9.12 (Hadamard's determinant inequality). Let $\mathbf{A} = (a_{ij}) = \mathbf{A}' \succ \mathbf{O}$. Prove

$$\det(\mathbf{A}) \leq \prod_{i=1}^n a_{ii},$$

with equality exactly when $\mathbf{A}$ is diagonal. Deduce that, if $\mathbf{r}'_i$ is row i of $\mathbf{M} \in \mathbb{R}^{n \times n}$, then

$$|\det(\mathbf{M})| \leq \prod_{i=1}^n \|\mathbf{r}_i\|.$$

If $\mathbf{M}$ is nonsingular, identify the equality condition.

Origin: A&M Ex. 12.26; S Prop. 6.5 and Thm. 6.9

Solution. Let $\mathbf{A} = \mathbf{L}\mathbf{L}'$ be its positive-diagonal Cholesky factorisation. Then

$$a_{ii} = \sum_{j=1}^i \ell_{ij}^2 \geq \ell_{ii}^2, \quad \det(\mathbf{A}) = \prod_{i=1}^n \ell_{ii}^2 \leq \prod_{i=1}^n a_{ii}.$$

All factors are positive; equality holds if and only if

$$a_{ii} = \ell_{ii}^2 \quad (i = 1, \dots, n) \iff \ell_{ij} = 0 \quad (j < i) \iff \mathbf{L} \text{ is diagonal} \iff \mathbf{A} \text{ is diagonal.}$$

If $\det(\mathbf{M}) = 0$, the row inequality holds. If $\mathbf{M}$ is nonsingular, $\mathbf{M}\mathbf{M}' \succ \mathbf{O}$, so

$$|\det(\mathbf{M})|^2 = \det(\mathbf{M}\mathbf{M}') \leq \prod_{i=1}^n (\mathbf{M}\mathbf{M}')_{ii} = \prod_{i=1}^n \|\mathbf{r}_i\|^2.$$

The nonnegative square roots give the inequality. Equality holds exactly when $\mathbf{M}\mathbf{M}'$ is diagonal, equivalently when the rows of $\mathbf{M}$ are pairwise orthogonal. ■

Exercise 9.13 (Residual orthogonality). Use the notation and assumptions of Theorem 9.23. Prove

$$\text{cov}(\mathbf{R}_\mathbf{B}, \mathbf{X}) = \mathbf{O} \iff \mathbf{B} = \mathbf{B}_*.$$

For $\mathbf{B} = \mathbf{B}_*$, decompose $\mathbf{Y} - \boldsymbol{\mu}_\mathbf{Y}$ into two vectors with zero cross-covariance.

Origin: Course formulation from H Cor. 14.8.6

Solution. Covariance calculus gives

$$\text{cov}(\mathbf{R}_B, \mathbf{X}) = \Sigma_{YX} - \mathbf{B}\Sigma_{XX}.$$

Since Σ_{XX} is nonsingular, this matrix is zero exactly when $\mathbf{B} = \Sigma_{YX}\Sigma_{XX}^{-1} = \mathbf{B}_*$. With $\mathbf{R}_* = \mathbf{R}_{B_*}$,

$$\mathbf{Y} - \mu_Y = \mathbf{B}_*(\mathbf{X} - \mu_X) + \mathbf{R}_*,$$

and

$$\text{cov}(\mathbf{B}_*(\mathbf{X} - \mu_X), \mathbf{R}_*) = \mathbf{B}_* \text{cov}(\mathbf{X}, \mathbf{R}_*) = \mathbf{0}.$$

Exercise 9.14 (A constrained linear maximum). Let $\mathbf{A} \succ \mathbf{0}$ and $\mathbf{a} \in \mathbb{R}^n$. Prove

$$\max_{\mathbf{x}'\mathbf{A}\mathbf{x} \leq 1} \mathbf{a}'\mathbf{x} = \sqrt{\mathbf{a}'\mathbf{A}^{-1}\mathbf{a}}.$$

For $\mathbf{a} \neq \mathbf{0}$, identify the unique maximiser.

Origin: A&M Ex. 12.1; H Thm. 14.10.1

Solution. Set $\mathbf{y} = \mathbf{A}^{1/2}\mathbf{x}$ and $\mathbf{c} = \mathbf{A}^{-1/2}\mathbf{a}$. Then

$$\mathbf{x}'\mathbf{A}\mathbf{x} = \|\mathbf{y}\|^2, \quad \mathbf{a}'\mathbf{x} = \mathbf{c}'\mathbf{y} \leq \|\mathbf{c}\|\|\mathbf{y}\| \leq \|\mathbf{c}\|.$$

Moreover,

$$\|\mathbf{c}\|^2 = \mathbf{a}'\mathbf{A}^{-1}\mathbf{a}.$$

If $\mathbf{a} \neq \mathbf{0}$, equality in both inequalities requires $\mathbf{y} = \mathbf{c}/\|\mathbf{c}\|$, hence the unique maximiser is

$$\mathbf{x}_* = \frac{\mathbf{A}^{-1}\mathbf{a}}{\sqrt{\mathbf{a}'\mathbf{A}^{-1}\mathbf{a}}}.$$

If $\mathbf{a} = \mathbf{0}$, the maximum is zero at every feasible point. ■

Exercise 9.15 (Mahalanobis distance after a rank-one update). Let $\Sigma \succ \mathbf{O}$, $\mathbf{u}, \mathbf{x} \in \mathbb{R}^p$, and $\tilde{\Sigma} = \Sigma + \mathbf{u}\mathbf{u}'$. Prove

$$\mathbf{x}'\tilde{\Sigma}^{-1}\mathbf{x} = \mathbf{x}'\Sigma^{-1}\mathbf{x} - \frac{(\mathbf{x}'\Sigma^{-1}\mathbf{u})^2}{1 + \mathbf{u}'\Sigma^{-1}\mathbf{u}} \leq \mathbf{x}'\Sigma^{-1}\mathbf{x}.$$

Give the equality condition.

Origin: Course formulation from H Thm. 18.2.8 and Thm. 18.3.4; A&M Ex. 12.16

Solution. The rank-one inverse identity gives

$$\tilde{\Sigma}^{-1} = \Sigma^{-1} - \frac{\Sigma^{-1}\mathbf{u}\mathbf{u}'\Sigma^{-1}}{1 + \mathbf{u}'\Sigma^{-1}\mathbf{u}}.$$

Premultiplication and postmultiplication by $\mathbf{x}'$ and $\mathbf{x}$ give the displayed equality. Since

$$1 + \mathbf{u}'\Sigma^{-1}\mathbf{u} > 0,$$

the subtracted term is nonnegative. Equality holds exactly when $\mathbf{x}'\Sigma^{-1}\mathbf{u} = 0$. 